

Souvik Das

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Professional Summary

Detail-oriented Metallurgy Engineer with **02 years of experience** in material testing, heat treatment, alloy development, and quality assurance in steel, automotive, and manufacturing industries. Skilled in analyzing material properties, failure investigations, and optimizing metallurgical processes to enhance performance, reduce costs, and ensure compliance with international standards (ASTM, ISO). Strong problem-solving abilities with a proven record of improving production efficiency and material strength.

Education

B.Tech in Metallurgical & Materials Engineering

IIT (BHU) Varanasi – 2024

- Key Subjects: Physical Metallurgy, Extractive Metallurgy, Welding, Material Science, Heat Treatment

Internship

Graduate Engineer Trainee (GET) – Metallurgy Intern

ESL | [06/2024] – [02/2025]

- Assisted in material testing (tensile, impact, fatigue, hardness) and documented results for ISO audits.
- Gained hands-on experience in foundry operations, casting inspection, and welding metallurgy.
- Supported senior engineers in process optimization projects, improving yield by **8%**.
- Prepared detailed metallurgical reports and presented findings to management.

Core Competencies

- Metallurgical Testing & Analysis (Microstructure, Hardness, Tensile, Impact)
- Heat Treatment & Surface Engineering
- Welding Metallurgy & Failure Analysis
- Alloy Design & Development
- Foundry Operations & Casting Metallurgy
- Quality Control & Non-Destructive Testing (NDT)
- Standards: ASTM, IS, ISO, ASME
- Software: AutoCAD, MATLAB, ANSYS, Solid Works, Thermo-Calc
- Root Cause Analysis (RCA), Six Sigma, Lean Manufacturing

Projects & Research

- **Failure Analysis of Heat-Treated Alloy Steel Components** – Conducted microstructure analysis to identify root causes of cracking.
 - **Development of High-Strength Aluminum Alloy** – Designed alloy with improved corrosion resistance for automotive use.
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Achievements

- Awarded “Best Project Engineer” for process optimization in steel plant.
 - Reduced raw material wastage by **10%** through improved process controls.
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Professional Experiences

Metallurgy Engineer

Hal off Store–Haldia Petrochemical LTD [03/2025] – [04/2026]

- Conducted metallurgical failure analysis of components, improving product reliability by **20%**.
- Designed and optimized heat treatment processes for steel/alloy components, reducing cycle time by **15%**.
- Performed microstructural analysis and hardness testing, ensuring compliance with ASTM standards.
- Collaborated with R&D to develop new alloy compositions, enhancing wear resistance in automotive parts.
- Implemented quality assurance procedures, decreasing rejection rate by **12%**.

Metallurgy Engineer

Mehta Tubes Ltd – Gujarat [06/2026] – Present

- Mehta Tubes Ltd (Copper Pipe Manufacturing)
- Oversee metallurgical and quality control activities in the manufacturing of copper tubes and pipes.
- Conduct mechanical and metallurgical testing, including hardness testing, dimensional inspection, and microstructural evaluation.
- Monitor annealing and heat treatment processes to achieve required mechanical properties and conductivity standards.
- Perform raw material inspection and chemical composition verification of copper and copper alloy products.
- Analyze manufacturing defects and implement corrective actions to improve product quality and reduce rejection rates.
- Coordinate with production and quality teams to optimize process parameters and enhance manufacturing efficiency.
- Ensure compliance with ASTM, IS, and customer-specific quality requirements.
- Prepare technical reports, inspection records, and root cause analysis (RCA) documentation.